Individual Diagnostic Requirement Specification

CN200S ABS

Version: V1.0

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Requirement wording conventions

Within this document the following conventions are applied when describing specific compliance requirements:

The word “Should” denotes a preference or desired conformance.

The word “Shall” is used in the following ways:

1. To state a binding requirement on the special network, or the ECUs which comprise the network, which is verifiable by external manipulation and/or observation of an ECU or the network.

2. To state a binding requirement upon an ECU’s requirements document that is verifiable through a review of the document.

The word “Must” is used to state a binding requirement upon ECUs on the network which have a corresponding ECU requirements document. These requirements will be verified as part of the component verification.

NOTE: In the event of a conflict between the text of this specification and the documents cited herein, the text of this specification takes precedence.

NOTE: Nothing in this specification supersedes applicable laws and regulations unless a specific exemption has been obtained.

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History record of the document

The following table contains the revision history for this specification.
Changes will not be listed if corrections of the spelling or formal changes are made.

Version#	Date	Editor	Description of the revision
V1.0	2014-05-20	Liao wenqing	Initial release

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1 Introduction

1.1 Scope

The detailed description about this chapter can be found in chapter 1.1 of the document *CN200S General Diagnostic Requirement Specification*.

NOTE: For CN200S vehicle, ABS needn't to realize the software download function, but ABS shall comply with the requirements for the non-programmable ECU, which is defined in chapter 16 of the document *CN200S General Diagnostic Requirement Specification*.

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2 Normative references

This document references ISO, SGMW and other publications as listed in the following sub sections.

2.1 ISO Publications

ISO 11898-1:2003 *Road Vehicles - Controller Area Networks (CAN) - Part 1: Data link layer and physical signalling*

ISO 11898-2:2003 *Road Vehicles - Controller Area Networks (CAN) - Part 2: High-speed medium access unit*

ISO 15765-1:2004 *Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 1: General information*

ISO 15765-2:2004 *Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 2: Network layer services*

ISO 15765-3:2004 *Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 3: Implementation of unified diagnostic services (UDS on CAN)*

ISO 14229-1:2006 *Road Vehicles - Unified diagnostic services (UDS) - Specification and requirements*

ISO 15031-6:2005 *Road Vehicles - Communication between vehicle and external equipment for emission-related diagnostic - Part 6: Diagnostic trouble code definitions*

2.2 SGMW Standard

CN200S General Diagnostic Requirement Specification

BT/SGMWJ 0858:2013 高速 CAN 物理层规范

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3 Terms, definitions and abbreviations

3.1 ISO Terms

This document makes use of terms defined in the ISO 15765-2, ISO 15765-3 and ISO 14229-1.

3.2 Specific Terms Defined By This Document

Table 3-1Specific Terms Defined By This Document

Abbreviation/缩写	Description/描述
msec	**mili**sec**ond**
CANId	**CAN I**dentifier
ABS/ESP	**A**nti-lock **B**rake **S**ystem/**E**lectronic **S**tability **P**rogram
TPMS	**T**ire **P**ressure **M**onitor **S**ystem
SDM	**R**estraint **S**ensing and**D**iagnostic **M**odule
DCU	**D**omain **C**ontroller **U**nit
BCM	**B**ody **C**ontrol **M**odule
PEPS	**P**assive **E**ntry**&P**assive **S**tart
EPS	**E**lectronic **P**ower **S**teering
AC	**Auto fixed-temp AC**
IC	**I**nstrument **C**luster
ICE	**I**n **C**ar **E**ntertainment
ECM	**E**ngine **C**ontrol **M**odule
TCM	**T**ransmission **C**ontrol **M**odule

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4 General requirement

All the general requirements, with which the tester and ABS shall comply, can be found in section 4 of the document *CN200S General Diagnostic Requirement Specification*.

4.1 Diagnostic CAN identifiers definition

The following table provides an overview about the definition of ABS diagnostic CAN identifier.

Table 4-1 ABS diagnostic CAN identifier

ECU		Functionally addressed request from tester to ECUs	Physically addressed request from tester to ECU	Response from ECU to tester
Name	Emission-related OBDECU (YES/NO)			
ABS	NO	0x7DF	0x720	0x728

4.2 ECU Diagnostic Address definition

The following table provides an overview about the definition of ABS diagnostic Address.

Table 4-2 ABS Diagnostic Address

ECU Name	Description	Value	Comments
ABS	ECUDiagnosticAddress	0x29	ABS diagnostic address, which is defined in *CN200S General Diagnostic Requirement Specification*.

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5 Physical layer requirement

All the physical layer requirements, with which the tester and ABS shall comply, can be found in section 5 of the document *CN200S General Diagnostic Requirement Specification.*

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6 Data link layer requirement

All the data link layer requirements, with which the tester and ABS shall comply, can be found in section 6 of the document *CN200S General Diagnostic Requirement Specification.*

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7 Network layer requirement

All the network layer requirements, with which the tester and ABS shall comply, can be found in section 7 of the document *CN200S General Diagnostic Requirement Specification.*

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8 Application and session layers requirement

All the application and session layer requirements, with which the tester and ABS shall comply, can be found in section 8 of the document *CN200S General Diagnostic Requirement Specification*.

8.1 Diagnostic Service Table Overview

The table followed (Table 8-1) shows the services which are allowed during the defaultSession, programmingSession and extendedSession of ABS. ABS is allowed to support more other diagnostic services per session after confirmed by SGMW.

Table 8-1 Diagnostic Service Table Overview

Service Name 服务名称	SID (Hex)	Sub-function/ 子功能 (Hex)	Session/会话						Security Access[1] / 安全访问		
			Default 默认		Programming 编程		Extended 扩展		LV1 等级 1	LV 2 等级 2	LV 3 等级 3
			Phys 物理寻址	Func 功能寻址	Phys 物理寻址	Func 功能寻址	Phys 物理寻址	Func 功能寻址			
Diagnostic and communication management functional unit/诊断与通信管理功能单元											
DiagnosticSessionControl	10	01	√[2]	√	-	-	√	√	×[2]	×	×
		-	-	-	-	-	-	-	-	-	-
		03	√	√	-	-	√	√	×	×	×
ECUReset	11	01	√	√	-	-	√	√	×	×	×
		02	√	√	-	-	√	√	×	×	×
SecurityAccess	27	01/02	×	×	-	-	√	×	×	×	×
		-	-	-	-	-	-	-	-	-	-
		-	-	-	-	-	-	-	-	-	-
CommunicationControl	28	00/80/03/83	×	×	-	-	√	√	×	×	×
TesterPresent	3E	00/80	√	√	-	-	√	√	×	×	×
ControlDTCSetting	85	01/02	×	×	-	-	√	√	×	×	×
ProgrammingMode	A5	01/03	×	×	-	-	√	√	×	×	×
Data transmission functional unit/数据传输功能单元											
ReadDataByIdentifier	22	-	√	√[3]	-	-	√	√[3]	×	×	×
DynamicallyDefineDataIdentifier	2C	-	-	-	-	-	-	-	-	-	-
WriteDataByIdentifier	2E	-	×	×	-	-	√	×	√[4]	×	×
Stored data transmission functional unit/存储数据传输功能单元											
ClearDiagnosticInformation	14	-	√	√	-	-	√	√	×	×	×
ReadDTCInformation	19	01/02/03/04/06/0A	√	√	-	-	√	√	×	×	×
InputOutput control functional unit/输入输出控制功能单元											
InputOutputControlByIdentifier	2F	-	-	-	-	-	-	-	-	-	-

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Service Name 服务名称	SID (Hex)	Sub-function/ 子功能 (Hex)	Session/会话						Security Access[1] / 安全访问		
			Default 默认		Programming 编程		Extended 扩展		LV1 等级 1	LV 2 等级 2	LV 3 等级 3
			Phys 物理寻址	Func 功能寻址	Phys 物理寻址	Func 功能寻址	Phys 物理寻址	Func 功能寻址			
Remote activation of routine functional unit/程序远程激活功能单元											
RoutineControl	31	01/02/03	×	×	-	-	√	×	×	×	×
Upload download functional unit/上传下载功能单元											
RequestDownload	34	-	-	-	-	-	-	-	-	-	-
TransferData	36	-	-	-	-	-	-	-	-	-	-
RequestTransferExit	37	-	-	-	-	-	-	-	-	-	-

NOTE:
注：

1. The definition of the security access level can be found in chapter 错误！未找到引用源。 of this specification.
   关安全访问等级的定义可参见本规范中的第错误！未找到引用源。章节。
2. ‘√’: Support; ‘×’: Not support, ‘-’: Not Applicable.(N/A)
   “√”：支持；“×”：不支持；“-”不适用。
3. If the ReadDataByIdentifier (22 hex) service request message is a single frame, the functional addressing method shall be supported; otherwise, the functional addressing method is not required to be supported.
   如果通过标识符读数据（22 hex）服务请求消息为单帧，则应（shall）支持功能寻址，否则，不需要支持功能寻址。
4. All the WriteDataByIdentifier (2E hex) service in all ECUs shall need the security access, and the security level needed by this service depends on each DID(see annex C)
   所有 ECU 的通过标识符写数据（2E hex）服务均应（shall）需要安全访问，该服务所需的安全等级取决于具体 DID（见附录 C）。

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9 Diagnostic service documentation structure

All the requirements about diagnostic service documentation structure, with which the tester and ABS shall comply, can be found in section 9 of the document *CN200S General Diagnostic Requirement Specification.*

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10 Diagnostic and communication management functional unit

10.1 DiagnosticSessionControl (10 hex)

10.1.1 Service description

All general requirements about the service can be found in related section of CN200S General Diagnostic Requirement Specification.

10.1.2 Request message

Table 10-1 Request message definition

A_Data Byte	Parameter Name	Hex value
#1	DiagnosticSessionControl Request Service Id	10
#2	DiagnosticSessionType=[DefaultSession ExtendedDiagnosticSession]	 01 03

10.1.3 Positive response message

Table 10-2 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	DiagnosticSessionControl Response Service Id	50
#2	DiagnosticSessionType=[DefaultSession ExtendedDiagnosticSession]	 01 03
 #3 #4 #5 #6	sessionParameterRecord[]#1=[$P2_{CAN_ECU_max}$(high byte) $P2_{CAN_ECU_max}$(low byte) $P2^*_{CAN_ECU_max}$(high byte) $P2^*_{CAN_ECU_max}$(low byte)]	 00 32 01 F4

NOTE: The definition about parameter $P2_{CAN_ECU_max}$ and $P2^*_{CAN_ECU_max}$ can be found in chapter 8.2.2 of the document *CN200S General Diagnostic Requirement Specification* and ISO 15765-3. These two parameters in positive response message are the ones that are used by ABS internally, if the value of this two parameters which is used by ABS are different from the definition in this document, it should be determined by SGMW and ABS Supplier together.

10.1.4 Supported negative response codes

Table 10-3 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect**

NOTE: In addition to the NRC which listed in the table above, ABS should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification*.

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10.2 ECUReset (11 hex)

10.2.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

10.2.2 Request message

Table 10-4 Request message definition

A_Data Byte	Parameter name	Hex value
#1	ECUReset Request Service Id	11
#2	resetType =[hardReset keyOffOnReset]	 01 02

10.2.3 Positive response message

Table 10-5 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	ECUReset Response Id	51
#2	resetType =[hardReset keyOffOnReset]	 01 02

10.2.4 Supported negative response codes

Table 10-6 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**

NOTE: In addition to the NRC which listed in the table above, ABS should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification*.

10.3 SecurityAccess (27 hex)

10.3.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

NOTE: This service, as well as MEC provides a protected mechanism to access the diagnostic services. The detailed definition about MEC can be found in chapter 16.1.3.2.5 of the document *CN200S General Diagnostic Requirement Specification*..

NOTE: The MEC mechanism is only applicable to Level1 of security access service (27 hex).
注：MEC机制只适用于安全访问服务的安全等级Level1.

10.3.2 Request message

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Table 10-7 Request message definition

A_DataByte	Parameter Name	Hex value
#1	SecurityAccess Request Service Id	27
#2	securityAccessType=requestSeed	01

Table 10-8 Request message definition

A_DataByte	Parameter Name	Hex value
#1	SecurityAccess Request Service Id	27
#2	securityAccessType =sendKey	02
	securityKey[]=[	
#3	key #1(MSB)	00-FF
#4	key #2	:
#5	key #3	:
#6	key#4(LSB)]	00-FF

10.3.3 Positive response message

Table 10-9 Positive response message definition (sub-function = requestSeed)

A_DataByte	Parameter Name	Hex value
#1	Security Access Response Service Id	67
#2	securityAccessType=requestSeed	01
	securitySeed[]=[	
#3	seed#1(MSB)	00-FF
#4	seed#2	:
#5	seed#3	:
#6	seed#4(LSB)]	00-FF

Table 10-10 Positive response message definition (sub-function = sendKey)

A_DataByte	Parameter Name	Hex value
#1	Security Access Response Service Id	67
#2	securityAccessType =sendKey	02

10.3.4 Supported negative response codes

Table 10-11 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect**
24	**requestSequenceError**
35	**invalidKey**
36	**exceededNumberOfAttempts**
37	**requireTimeDelayNotExpired**

NOTE: In addition to the NRC which listed in the table above, ABS should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification*.

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10.4 CommunicationControl (28 hex)

N/A

10.5 TesterPresent (3E hex)

10.5.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

10.5.2 Request message

Table 10-12 Request message definition

A_DataByte	Parameter Name	Hex value
#1	TesterPresent Request Service Id	3E
#2	zeroSubFunction=[zeroSubFunction (suppressPosRspMsgIndicationBit =0) zeroSubFunction (suppressPosRspMsgIndicationBit =1)]	 00 80

10.5.3 Positive response message

Table 10-13 Positive response message definition(sub-function = zeroSubFunction (suppressPosRspMsgIndicationBit =0))

A_DataByte	Parameter Name	Hex value
#1	TesterPresent Response Service Id	7E
#2	zeroSubFunction= zeroSubFunction (suppressPosRspMsgIndicationBit =0)	00

sub-function = zeroSubFunction (suppressPosRspMsgIndicationBit =1)

NOTE: There is no any response for the request message, because of suppressPosRspMsgIndicationBit =1.

10.5.4 Supported negative response codes

Table 10-14 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**

NOTE: In addition to the NRC which listed in the table above, ABS should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification*.

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10.6 AccessTimingParameter (83 hex)

N/A

10.7 SecuredDataTransmission (84 hex)

N/A

10.8 ControlDTCSetting (85 hex)

N/A

10.9 ResponseOnEvent (86 hex)

N/A

10.10 LinkControl (87 hex)

N/A

10.11 ProgrammingMode (A5 hex)

N/A

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11 Data transmission functional unit

11.1 ReadDataByIdentifier (22 hex)

11.1.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

NOTE: The maximum number of dataIdentifier that can be simultaneously requested is five.

NOTE: The DID which used to identify the general data record for all the ECUs called vehicle-manufacturer-specific DID, the definition about vehicle-manufacturer-specific DID can be found in chapter C.1 of Annex C. The DID which used to identify the individual data record for each ECU called ECU-supplier-specific DID, the definition about ECU-supplier-specific DID can be found in in chapter C.2 of Annex C.

11.1.2 Request message

Table 11-1 Request message definition

A_DataByte	Parameter Name	Hex value
#1	ReadDataByIdentifier Request Service Id	22
	dataIdentifier[]#1=[	
#2	byte #1(MSB)	00-FF
#3	byte #2(LSB)]	00-FF
:	:	:
	dataIdentifier[]#m[a]=[	
#n-1	byte #1(MSB)	00-FF
#n	byte #2(LSB)]	00-FF
a. m≤5		

11.1.3 Positive response message

Table 11-2 Positive response message definition

A_DataByte	Parameter Name	Hex value
#1	ReadDataByIdentifier Response Service Id	62
	dataIdentifier[]#1=[	
#2	byte #1(MSB)	00-FF
#3	byte #2(LSB)]	00-FF
	dataRecord[]#1=[	
#4	data #1	00-FF
:	:	:
#(k-1)+4	data #k [a]]	00-FF
:	:	:
	dataIdentifier[]#m[b]=[	
#n-(o-1)-2	byte #1(MSB)	00-FF
#n-(o-1)-1	byte #2(LSB)]	00-FF
	dataRecord[]#m[b]=[	
#n-(o-1)	data #1	00-FF

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: #n	: data #o [a]]	: 00-FF
a. The number of data# depends on specified dataIdentifier in chapter C.1 and chapter C.2 of Annex C. b. m≤5		

11.1.4 Supported negative response codes

Table 11-3 Supported negative response codes

Hex	Description
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect** *The corresponding correct condition for ABS to implement the request message shall be defined by the ABS supplier.*

NOTE: In addition to the NRC which listed in the table above, ABS should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification*.

11.2 ReadMemoryByAddress (23 hex)

N/A

11.3 ReadScalingDataByIdentifier (24 hex)

N/A

11.4 ReadDataByPeriodicIdentifier (2A hex)

N/A

11.5 DynamicallyDefineDataIdentifier (2C hex)

N/A

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11.6 WriteDataByIdentifier (2E hex)

11.6.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

11.6.2 Request message

Table 11-4 Request message definition

A_DataByte	Parameter Name	Hex value
#1	WriteDataByIdentifier Request Service Id	2E
	dataIdentifier[]=[	
#2	byte #1(MSB)	00-FF
#3	byte #2(LSB)]	00-FF
	dataRecord[]=[	
#4	data #1	00-FF
:	:	:
#m+3	data #m[a]]	00-FF

a. The content of data# depends on specified dataIdentifier in chapter C.1 and chapter C.2 of Annex C.

11.6.3 Positive response message

Table 11-5 Positive response message definition

A_DataByte	Parameter Name	Hex value
#1	WriteDataByIdentifier Response Service Id	6E
	dataIdentifier[]=[	
#2	byte #1(MSB)	00-FF
#3	byte #2(LSB)]	00-FF

11.6.4 Supported negative response codes

Table 11-6 Supported negative response codes

Hex	Description
33	**securityAccessDenied**
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**
72	**generalProgrammingFailure**

NOTE: In addition to the NRC which listed in the table above, ABS should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification*.

11.7 WriteMemoryByAddress (3D hex)

N/A

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12 Stored data transmission functional unit

12.1 ClearDiagnosticInformation (14 hex)

12.1.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

12.1.2 Request message

Table 12-1 Request message definition

A_DataByte	Parameter Name	Hex value
#1	ClearDiagnosticInformation Request Service Id	14
	groupOfDTC[]=[	
#2	groupOfDTCHighByte	FF
#3	groupOfDTCMiddleByte	FF
#4	groupOfDTCLowByte]	FF

12.1.3 Positive response message

Table 12-2 Positive response message definition

A_DataByte	Parameter Name	Hex value
#1	ClearDiagnosticInformation Response Service Id	54

12.1.4 Supported negative response codes

Table 12-3 Supported negative response codes

Hex	Description
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**

NOTE: In addition to the NRC which listed in the table above, ABS should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification*.

12.2 ReadDTCInformation (19 hex)

12.2.1 Service description

12.2.2 Request message

Table 12-4 Request message definition (sub-function = reportNumberOfDTCByStatusMask)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Request Service Id	19
#2	reportType=[	
	reportNumberOfDTCByStatusMask	01
	reportDTCByStatusMask]	02
#3	DTCStatusMask	00-FF

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Table 12-5 Request message definition (sub-function = reportDTCSnapshotRecordByDTCNumber)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Request Service Id	19
#2	reportType=reportDTCSnapshotRecordByDTCNumber	04
#3 #4 #5	DTCMaskRecord[]=[DTCHighByte DTCMiddleByte DTCLowByte]	 00-FF 00-FF 00-FF
#6	DTCSnapshotRecordNumber	FF[a]
a. A value of FF hex requests the ECU to report all stored DTCSnapshot data records at once.		

Table 12-6 Request message definition (sub-function = reportDTCExtendedDataRecordByDTCNumber)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Request Service Id	19
#2	reportType=reportDTCExtendedDataRecordByDTCNumber	06
#3 #4 #5	DTCMaskRecord[]=[DTCHighByte DTCMiddleByte DTCLowByte]	 00-FF 00-FF 00-FF
#6	DTCExtendedDataRecordNumber	00-FF[a]
a. A value of FF hex requests the ECU to report all stored DTCExtendedData records in a single response message.		

Table 12-7 Request message definition (Sub-function = reportSupportedDTC)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Request Service Id	19
#2	reportType=reportSupportedDTC	0A

12.2.3 Positive response message

Table 12-8 Positive response message definition (sub-function = reportNumberOfDTCByStatusMask)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59
#2	reportType=reportNumberOfDTCByStatusMask	01
#3	DTCStatusAvailabilityMask	AB
#4	ISO 15031-6 DTCFormat	00
#5 #6	DTCCount[]=[DTCCountHighByte DTCCountLowByte]	 00-FF 00-FF

Table 12-9 Positive response message definition (sub-function = reportDTCByStatusMask, reportSupportedDTC)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59
#2	reportType=[reportDTCByStatusMask reportSupportedDTC]	 02 0A
#3	DTCStatusAvailabilityMask	AB

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	DTCAndStatusRecord[] = [	
#4	DTCHighByte #1	00-FF
#5	DTCMiddleByte #1	00-FF
#6	DTCLowByte #1	00-FF
#7	statusOfDTC #1	00-FF
#8	DTCHighByte #2	00-FF
#9	DTCMiddleByte #2	00-FF
#10	DTCLowByte #2	00-FF
#11	statusOfDTC #2	00-FF
:	:	:
#n-3	DTCHighByte #m	00-FF
#n-2	DTCMiddleByte #m	00-FF
#n-1	DTCLowByte #m	00-FF
#n	statusOfDTC #m]	00-FF

NOTE: If reportType = reportSupportedDTC:
1. ABS should report the status of all the DTC's which is supported by ABS.
2. ABS should report the status at the time of receiving the requirement.

Table 12-10 Positive response message definition (sub-function = reportDTCSnapshotRecordByDTCNumber)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59
#2	reportType=reportDTCSnapshotRecordByDTCNumber	04
	DTCAndStatusRecord[]=[	
#3	DTCHighByte	00-FF
#4	DTCMiddleByte	00-FF
#5	DTCLowByte	00-FF
#6	statusOfDTC]	00-FF
#7	DTCSnapshotRecordNumber#1	01
#8	DTCSnapshotRecordNumberOfIdentifiers#1	03
	DTCSnapshotRecord[]#1=[	
#9	ECU Voltage DataIdentifier(MSB)[a]	F0
#10	ECU Voltage DataIdentifier(LSB)[a]	11
#11	ECU Voltage[a]	00-FF
#12	Engine Speed DataIdentifier(MSB)[a]	F0
#13	Engine Speed DataIdentifier(LSB)[a]	12
#14	Engine Speed (High Byte)[a]	FF
#15	Engine Speed (Low Byte)[a]	FF
#16	Vehicle Speed DataIdentifier(MSB)[a]	F0
#17	Vehicle Speed DataIdentifier(LSB)[a]	13
#18	Vehicle Speed (High Byte)[a]	00-FF
#19	Vehicle Speed (Low Byte)[a]	00-FF

a. This part is the general Snapshot for all ECUs, detailed definitions of general snapshot record can be found in chapter 12.2.3.2 of *CN200S General Diagnostic Requirement Specification*.

NOTE: For ABS, only ECU Voltage and Vehicle Speed are supported, for Engine Speed that ABS does not supported, the default value should be 0xFFFF.

Table 12-11 Positive response message definition (sub-function = reportDTCExtendedDataRecordByDTCNumber)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59

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#2	reportType=reportDTCExtendedDataRecordByDTCNumber	06
#3 #4 #5 #6	DTCAndStatusRecord[]=[DTCHighByte DTCMiddleByte DTCLowByte statusOfDTC]	 00-FF 00-FF 00-FF 00-FF
#7	DTCExtendedDataRecordNumber #1[a]	01
#8	Fault occurrence counter[a]	00-FF
#9	DTCExtendedDataRecordNumber #2[a]	02
#10	Fault pending counter[a]	FF
#11	DTCExtendedDataRecordNumber #3[a]	03
#12	DTC Aging counter[a]	FF
#13	DTCExtendedDataRecordNumber #4[a]	04
#14	DTC Aged counter[a]	FF

a. This part is the DTCExtendedDataRecord for all ECUs, detailed definitions of DTC extended data record can be found in chapter 12.2.3.2 of *CN200S General Diagnostic Requirement Specification*. Whether the parameter appeared or not and the number of these parameters that appeared depend on the specific service requested.

NOTE: For ABS, only Fault occurrence counter is supported, for DTC extended data that ABS does not supported, the default value should be 0xFF.

12.2.4 Supported negative response codes

Table 12-12 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**

NOTE: In addition to the NRC which listed in the table above, ABS should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification*.

13 InputOutput control functional unit

InputOutputControlByIdentifier (2F hex)

N/A

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14 Remote activation of routine functional unit

14.1 RoutineControl (31 hex)

14.1.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

NOTE: If the valves are activated for too long the ECU can be damaged. The maximum activation time of the valves must not exceed 30 seconds at an ambient temperature of 30°C. Using this service repeatedly can damage the valves or pump. Please keep this information in mind when using this service.

14.1.2 Request message

Table 14–1 Request message definition

A_Data Byte	Parameter Name	Hex value
#1	RoutineControl Request Service Id	31
#2	RoutineControlType=[startRoutine stopRoutine requestRoutineResults]	 01 02 03
 #3 #4	routineIdentifier[][a,b]=[byte#1(MSB) byte#2(LSB)]	 00-FF 00-FF
 #5 : #m+4	routineControlOptionRecord[][a,b,c]=[routineControlOption#1 : routineControlOption#m]	 00-FF : 00-FF

a. **If RoutineControlType = startRoutine:**
For ABS, whether the parameter **routineControlOptionRecord** is present or not depends on the RIDs which used by ABS, the detailed definitions can be found in Annex F.2 of Annex F;

b. **If RoutineControlType = stopRoutine:**
The parameter **routineControlOptionRecord** is **NOT** present.
For ABS, whether the RID can be stopped by routineControlType = Stop Control or not depends on the RIDs which used by ABS, the detailed definitions can be found in Annex F.2 of Annex F;

c. **If RoutineControlType = requestRoutineResults:**
the parameter **routineControlOptionRecord** is **NOT** present.

14.1.3 Positive response message

Table 14–2 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	RoutineControl Response Service Id	71
#2	RoutineControlType=[startRoutine stopRoutine requestRoutineResults]	 01 02 03
	routineIdentifier[]=[	

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#3 #4	byte#1 (MSB) byte#2 (LSB)]	00-FF 00-FF
#5 : #m+4	routineStatusRecord[][a]=[routineStatus #1 : routineStatus #m]	00-FF : 00-FF
a. The detailed definition about routineStatusRecord can be found in chapter F.3 of Annex F.		

14.1.4 Supported negative response codes

Table 14–3 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**

NOTE: In addition to the NRC which listed in the table above, ABS should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification*.

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15 Upload download functional unit

15.1 RequestDownload (34 hex)

N/A

15.2 RequestUpload (35 hex)

N/A

15.3 TransferData (36 hex)

N/A

15.4 RequestTransferExit (37 hex)

N/A

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16 ECU programming requirements and process

The detailed description about this chapter can be found in chapter 16 of the document *CN200S General Diagnostic Requirement Specification*.

NOTE: For CN200S vehicle, ABS needn't to realize the software download function, but ABS shall comply with the requirements for the non-programmable ECU, which is defined in chapter 16 of the document *CN200S General Diagnostic Requirement Specification*.

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Annex A (normative) Global parameter definitions

The detailed description about this Annex can be found in Annex A of the document *CN200S General Diagnostic Requirement Specification*.

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Annex B (normative) Diagnostic and communication management functional unit data parameter definitions

The detailed description about this Annex can be found in Annex B of the document *CN200S General Diagnostic Requirement Specification*.

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Annex C (normative) Data transmission functional unit data parameter definitions

C.1 Vehicle-manufacturer-specific DID definitions

C.2 ECU-supplier-specific DID definitions

The following table lists the definition about ECU-supplier-specific DID. According to the requirements of the document *CN200S General Diagnostic Requirement Specification*, the range which ABS used is 0x2001~0x2FFF.

DID (Hex)	Data Description	Supported by Service (Hex)		Data Length (Byte)	Data Type
		22	2E		
2D00	SpeedInformation	x	-	10	Hex
2D01	InputData	x	-	2	Hex
2D02	ActuationState	x	-	3	Hex&BIN
2D03	FillingInStatus	x	x^1	1	Hex
2D04	EOLStatus	x	x^1	1	Hex
2D07	Speed Limit Disable/Enable	x	-	1	Hex
2D08	Disable Valve Relay Shutdown	x	-	1	Hex
2D10	SystemName	x	-	3	ASCII

NOTE: "×": Support without security access; "x^n": Support with security access (A superscript number 'n' after 'x' indicates that the number of the security level); "-": Not Supported.

C.2.1 DID 0x2D00- SpeedInformation

Byte No.	Description	Unit	Scaling	HEX limits	Physical limits
data #1	Vehicle speed (high Byte)	km/h	0.05625	0x0000~0x120E	0~260
data #2	Vehicle speed (low Byte)				
data #3	Wheel speed FL (high Byte)	km/h	0.05625	0x0000~0x120E	0~260
data #4	Wheel speed FL (low Byte)				
data #5	Wheel speed FR (high Byte)	km/h	0.05625	0x0000~0x120E	0~260
data #6	Wheel speed FR (low Byte)				
data #7	Wheel speed RL (high Byte)	km/h	0.05625	0x0000~0x120E	0~260
data #8	Wheel speed RL (low Byte)				
data #9	Wheel speed RR (high Byte)	km/h	0.05625	0x0000~0x120E	0~260
data #10	Wheel speed RR (low Byte)				

NOTE: If there is failure at any of the wheel speed sensor, the corresponding vehicle speed/ wheel speed will be invalid and the output will be 0x7FFF.

C.2.2 DID 0x2D01- InputData

Byte No.	Description	Unit	Scaling	HEX limits	Physical limits
data #1	Battery Voltage	V	0.08	0x00~0xFF	0~20.4
Byte No.	**Description**	**Unit**	**Scaling**	**Comments**	

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Byte No.	Description	Unit	Scaling	HEX limits	Physical limits
data #2	Brake Light Switch	-	-	0x00: OFF 0x01: ON	

C.2.3 DID 0x2D02- ActuationState

Byte No.	Description	Data Type	Comments		
data #1	Valve Relay Status	Hex	0x00: OFF 0x01: ON		
data #2	Pump Motor Status	Hex	0x00: OFF 0x01: ON		
data #3	Valve Actuation State(high Byte)	BIN	bit7	Outlet Valve Rear Right (AVRR)	0 = inactive 1 = active
			bit6	Inlet Valve Rear Right (EVRR)	0 = inactive 1 = active
			bit5	Outlet Valve Rear Left(AVRL)	0 = inactive 1 = active
			bit4	Inlet Valve Rear Left(EVRL)	0 = inactive 1 = active
			bit3	Outlet Valve Front Right (AVFR)	0 = inactive 1 = active
			bit2	Inlet Valve Front Rear (EVFR)	0 = inactive 1 = active
			bit1	Outlet Valve Front Left(AVFL)	0 = inactive 1 = active
			bit0	Inlet Valve Front Left(EVFL)	0 = inactive 1 = active
NOTE: If the bit(s) is (are) not used, the default value for each bit is “0”.					

C.2.4 DID 0x2D03- FillingInStatus

Byte No.	Description	Data Type	Default Value	Comments
data #1	FillingInStatus	Hex	0xFF	0x00 : Filling-in not completed 0xAA : Filling-in completed and ok 0xEE : Filling-in completed and not ok
NOTE: The data byte of delivery state from the supplier is 0xFF.				

C.2.5 DID 0x2D04- EOLStatus

Byte No.	Description	Data Type	Default Value	Comments
data #1	EOLStatus	Hex	0xFF	0x00 : End-of-line test not completed 0xAA : End-of-line test completed and ok 0xEE : End-of-line test completed and not ok
NOTE: The data byte of delivery state from the supplier is 0xFF.				

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C.2.8 DID 0x2D07- Speed Limit Disable/Enable

Byte No.	Description	Data Type	Default Value	Comments
data #1	Speed Limit Disable/Enable	Hex	-	0x00: Speed Limit Disabled 0x01: Speed Limit Enabled

C.2.9 DID 0x2D08- Disable Valve Relay Shutdown

Byte No.	Description	Data Type	Default Value	Comments
data #1	Disable Valve Relay Shutdown	Hex	-	0x00: Valve Relay Shutdown Disabled 0x01: Valve Relay Shutdown Enabled

C.2.10 DID 0x2D10- SystemName

Byte No.	Description	Data Type	Default Value	Comments			
data #1	SystemName	ASCII	*To be defined by ABS Supplier*	ABS	0x41	ESP	0x45
data #2					0x42		0x53
data #3					0x53		0x50
NOTE: The default value for 0x2D10 should be written into ECU EEPROM by ABS Supplier.							

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Annex D (normative) Stored data transmission functional unit data parameter definitions

The detailed description about this Annex can be found in Annex D of the document *CN200S General Diagnostic Requirement Specification*.

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Annex E (normative) Input output control functional unit data parameter definitions

E.1 InputOutputControlParameter definitions

The detailed description about this Annex can be found in Annex E.1 of the document *CN200S General Diagnostic Requirement Specification*.

E.2 InputOutputControlByIdentifier (2F hex) controlOptionRecord definitions

N/A

E.3 InputOutputControlByIdentifier (2F hex) controlEnableMaskRecord definitions

N/A

E.4 InputOutputControlByIdentifier (2F hex) controlStatusRecord definitions

N/A

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Annex F (normative) Remote activation of routine functional unit data parameter definitions

F.1 RoutineIdentifier definition

The detailed description about this Annex can be found in Annex F.1 of the document *CN200 General Diagnostic Requirement Specification*.

F.2 RoutineControl (31 hex) routineControlOption definitions

The following table lists the definition about routine identifier, According to the requirements of the document *CN200 General Diagnostic Requirement Specification*, the range which ABS used is 0x0200~0xDFFF.

RID (Hex)	Description	Supported routineControlType (Hex)		
		01	02	03
0300	DelayedActuatorControl	x	x	x
0301	EvacAndFill	x	x	x
0302	RepairBleed	x	x	x
0303	ActuatorTest	x	x	x
0304	DynamicTest	x	x	x
0305	WheelSpeedSensorsTest	x	x	x
0403	Speed Limit Disable/Enable	x	-	x
0404	Disable Valve Relay Shutdown	x	-	x
NOTE: "×": Support without security access; "×^{n}": Support with security access (A superscript number 'n' after 'x' indicates that the number of the security level); "-": Not Supported.				

The following sections list the definition about routine control option.

F.2.1 RID 0x0300- DelayedActuatorControl

Byte No.	routineControlOptionRecord	Data Type	value
data #1	ValveStates(State 1)	Hex	bit7: Outlet Valve Rear Right bit6: Inlet Valve Rear Right bit5: Outlet Valve Rear Left bit4: Inlet Valve Rear Left bit3: Outlet Valve Front Right bit2: Inlet Valve Front Right bit1: Outlet Valve Front Left bit0: Inlet Valve Front Left **NOTE:** For each control the corresponding bit has to be set to 1 to switch the actuator on and to 0 to switch the actuator off.

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Byte No.	routineControlOptionRecord	Data Type	value
data #2	DriverStates(State 1)	Hex	bit7: Unused bit6: Pump bit5: Unused bit4: Unused bit3: Unused bit2: Unused bit1: Unused bit0: Unused **NOTE:** For each control the corresponding bit has to be set to 1 to switch the actuator on and to 0 to switch the actuator off.
data #3	Activation time	Hex	Resolution: 20ms/bit, offset: 0.
data #4	ValveStates(State 2)	Hex	Same as data#1
data #5	DriverStates(State 2)	Hex	Same as data#2

NOTE: This request is used to control multiple drivers simultaneously. For each control two states are defined: state 1 and state 2. When the request was received the selected controls are set in state 1. After the activation time they are set to state 2. In comparison to the Actuator Test this service allows the tester to actuate one or more valves for a specified time. This feature allows the tester to run the customer specified End Of Line routine sequence instead of Pre-defined Periodic Actuator Control (EvacAndFill, RepairBleed) sequence.
NOTE: A "Failure" response has one of the following reasons: At least one valve or one driver doesn't exist or the control is not possible at this time because of safety reasons.
NOTE: A control that has been started by a delay actuator control request can be stopped by a request with the ControlOption set to "Stop control" (Emergency Stop, 31h 02h F0h 00h).
NOTE: At each end of this service the tester has to send the following request to turn off all controls: 31h 01h F0h 00h 00h 00h 00h 00h 00h.

F.2.2 RID 0x0301- EvacAndFill

Byte No.	routineControlOptionRecord	Unit	Scaling	Offset	Data Type	value
data #1	Evac and Fill time	S	1s/bit	0	Hex	2~150
data #2	Time until start of pump motor	S	1s/bit	0	Hex	2~150
	NOTE: 0xFF= no pump motor					

NOTE: This request is used to fill and drain the hydraulic unit.
NOTE: Time has to be between 2s and 150s. If Time exceeds this limit, the ECU uses the maximum defined value. After evac and filling time, the pump motor will run for 2 additional seconds, if the delay time in byte 5 is not 0xFF.

F.2.3 RID 0x0302- RepairBleed

Byte No.	routineControlOptionRecord	Data Type	value
data #1	Repair Bleeding phase	Hex	0x01: phase1 0x02: phase2 0x03: phase3 0x04: phase4
data #2	Repetition counter	Hex	0x03: 3 loops 0x04: 4 loops 0x05: 5 loops

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Byte No.	routineControlOptionRecord	Data Type	value
NOTE: For parameter "Repair Bleeding phase", the repair bleeding function should always be executed in increasing phase sequence starting with Phase 1. A repetition of the same phase is allowed, but no phase shall be skipped. If this is not ensured, the pump might be damaged due to dry run. **NOTE:** For parameter "Repetition counter", this parameter defines the number of inner loop repetitions in phase 3 and 4. This parameter is always required, but only valid for Phase 3 and 4, for other phases, it is ignored.			

F.2.4 RID 0x0303- ActuatorTest

Byte No.	routineControlOptionRecord	Data Type	value
data #1	Actuator 1 Value(high byte)	Hex	0x0000=OFF 0xFFFF=ON
data #2	Actuator 1 Value(low byte)	Hex	
data #3	Actuator 1 Identifier(high byte)	Hex	0x0000 = No actuation 0x0022 = Pump motor relay 0x0030 = Inlet valve front left 0x0032 = Outlet valve front left 0x0034 = Inlet valve front right 0x0036 = Outlet valve front right 0x0038 = Inlet valve rear right 0x003A = Outlet valve rear right 0x003C = Inlet valve rear left 0x003E = Outlet valve rear left 0x004E= HSV2 (ESP only) 0x0050= HSV1 (ESP only) 0x0052= USV2 (ESP only) 0x0054= USV1 (ESP only) 0x0002 = ABS Safety Lamp 0x0004 = ESP Safety Lamp (ESP Only) 0x000E = EBD Safety Lamp
data #4	Actuator 1 Identifier(low byte)	Hex	
data #5	Actuator 2 Value(high byte)	Hex	Same as data#1 and data#2
data #6	Actuator 2 Value(low byte)	Hex	
data #7	Actuator 2 Identifier(high byte)	Hex	Same as data#3 and data#4
data #8	Actuator 2 Identifier(low byte)	Hex	
data #9	Activation time(high byte)	Hex	Resolution: 10ms/bit, offset: 0. range: 10ms~327s
data #10	Activation time(low byte)	Hex	
data #11	Actuator 3 Value(high byte)	Hex	Same as data#1 and data#2
data #12	Actuator 3 Value(low byte)	Hex	
data #13	Actuator 3 Identifier(high byte)	Hex	Same as data#3 and data#4
data #14	Actuator 3 Identifier(low byte)	Hex	
data #15	Actuator 4 Value(high byte)	Hex	Same as data#1 and data#2
data #16	Actuator 4 Value(low byte)	Hex	
data #17	Actuator 4 Identifier(high byte)	Hex	Same as data#3 and data#4
data #18	Actuator 4 Identifier(low byte)	Hex	
NOTE: This request is used to start Actuator Test. **NOTE:** The pump motor relay must not be actuated for times shorter than 20ms. Shorter actuations could damage the motor relay due to very high current during the starting phase of the motor. Strongly recommends only using an ActuationValue of 0% and 100% **NOTE:** A negative response has one of the following reasons: At least one valve or one driver doesn't exist or the control is not possible at this time because of safety reasons. **NOTE:** If the valve relay is out of order, the routine will be aborted.			

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F.2.5 RID 0x0304- DynamicTest

Byte No.	routineControlOptionRecord	Data Type	value
data #1	Actuator 1 Value(high byte)	Hex	0x0000=OFF 0xFFFF=ON
data #2	Actuator 1 Value(low byte)	Hex	
data #3	Actuator 1 Identifier(high byte)	Hex	0x0000 = No actuation 0x0022 = Pump motor relay 0x0030 = Inlet valve front left 0x0032 = Outlet valve front left 0x0034 = Inlet valve front right 0x0036 = Outlet valve front right 0x0038 = Inlet valve rear right 0x003A = Outlet valve rear right 0x003C = Inlet valve rear left 0x003E = Outlet valve rear left 0x004E= HSV2 (ESP only) 0x0050= HSV1 (ESP only) 0x0052= USV2 (ESP only) 0x0054= USV1 (ESP only) 0x0002 = ABS Safety Lamp 0x0004 = ESP Safety Lamp (ESP Only) 0x000E = EBD Safety Lamp
data #4	Actuator 1 Identifier(low byte)	Hex	
data #5	Actuator 2 Value(high byte)	Hex	Same as data#1 and data#2
data #6	Actuator 2 Value(low byte)	Hex	
data #7	Actuator 2 Identifier(high byte)	Hex	Same as data#3 and data#4
data #8	Actuator 2 Identifier(low byte)	Hex	
data #9	Activation time(high byte)	Hex	Resolution: 10ms/bit, offset: 0. range: 10ms~327s
data #10	Activation time(low byte)	Hex	
data #11	Actuator 3 Value(high byte)	Hex	Same as data#1 and data#2
data #12	Actuator 3 Value(low byte)	Hex	
data #13	Actuator 3 Identifier(high byte)	Hex	Same as data#3 and data#4
data #14	Actuator 3 Identifier(low byte)	Hex	
data #15	Actuator 4 Value(high byte)	Hex	Same as data#1 and data#2
data #16	Actuator 4 Value(low byte)	Hex	
data #17	Actuator 4 Identifier(high byte)	Hex	Same as data#3 and data#4
data #18	Actuator 4 Identifier(low byte)	Hex	
data #19	Delay Time(high byte)	Hex	Resolution: 10ms/bit, offset: 0. range: 10ms~327s
data #20	Delay Time(low byte)	Hex	
NOTE: This request is used to start Dynamic Test.			

F.2.6 RID 0x0305- WheelSpeedSensorsTest

Byte No.	routineControlOptionRecord	Data Type	value
data #1	Execution time (high byte)	Hex	Resolution: 10ms/bit, offset: 0.
data #2	Execution time (low byte)	Hex	
NOTE: This request is used to start Wheel Speed Sensors Test. **NOTE:** Execution time can't be 0, or else NRC13 will be returned.			

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F.2.8 RID 0x0403- Speed Limit Disable/Enable

Byte No.	routineControlOptionRecord	Data Type	value
data #1	Speed Limit Disable/Enable	Hex	0x00 = Disable speed Limit 0x01 = Enable Speed Limit
NOTE: This request is Disable/Enable the speed Limit (10kph for Extended Mode). **NOTE:** This Routine 0x0403 cannot be stopped by ControlOption= Stop Control.			

F.2.9 RID 0x0404- Disable Valve Relay Shut down

Byte No.	routineControlOptionRecord	Data Type	value
data #1	Disable Valve Relay Shutdown	Hex	0x00 = Disable Valve Relay Shutdown 0x01 = Re-enable Valve relay shutdown
NOTE: This Routine 0x0404 cannot be stopped by ControlOption= Stop Control.			

F.3 RoutineControl (31 hex) routineStatus definitions

The following sections list the definition about routine status.

F.3.1 RID 0x0300- DelayedActuatorControl

Byte No.	Description	Data Type	value
data #1	Supported Control /Routine Status	Hex	If routineControlType = startRoutine, then 0x01 = start successful 0x02 = already started 0x03 = not possible to start Else if routineControlType = stopRoutine, then 0x01 = stop successful 0x02 = already stopped or not start 0x03 = routine stopped abnormal Else if routineControlType = requestRoutineResults, then 0x01 = finished 0x02 = still running 0x03 = stopped or not start 0x04 = routine stopped abnormal

F.3.2 RID 0x0301- EvacAndFill

Byte No.	Description	Data Type	value
data #1	Supported Control /Routine Status	Hex	Same as the Comments in the table of F.3.1

F.3.3 RID 0x0302- RepairBleed

Byte No.	Description	Data Type	value

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Byte No.	Description	Data Type	value
data #1	Supported Control/ Routine Status	Hex	Same as the Comments in the table of F.3.1

F.3.4 RID 0x0303- ActuatorTest

Byte No.	Description	Data Type	value
data #1	Supported Control/ Routine Status	Hex	Same as the Comments in the table of F.3.1
data #2	Result, actuator 1(high byte)	Hex	*The description should be defined by the ABS supplier and confirmed by SGMW.*
data #3	Result, actuator 1(low byte)	Hex	
data #4	Result, actuator 2(high byte)	Hex	*The description should be defined by the ABS supplier and confirmed by SGMW.*
data #5	Result, actuator 2(low byte)	Hex	
data #6	Result, actuator 3(high byte)	Hex	*The description should be defined by the ABS supplier and confirmed by SGMW.*
data #7	Result, actuator 3(low byte)	Hex	
data #8	Result, actuator 4(high byte)	Hex	*The description should be defined by the ABS supplier and confirmed by SGMW.*
data #9	Result, actuator 4(low byte)	Hex	
NOTE: These bytes from data #2 to data #9 are only for RoutineControlType = requestRoutineResults and Supported Control / Routine Status = “finished”.			

F.3.5 RID 0x0304- DynamicTest

Byte No.	Description	Data Type	value
data #1	Supported Control/ Routine Status	Hex	Same as the Comments in the table of F.3.1
data #2	delta wheel speed FL (high byte)	Hex	Resolution: 1/64/bit, offset: 0. range:0~100m/s
data #3	delta wheel speed FL (low byte)	Hex	
data #4	delta wheel speed FR (high byte)	Hex	Resolution: 1/64/bit, offset: 0. range: 0~100m/s
data #5	delta wheel speed FR (low byte)	Hex	
data #6	delta wheel speed RL (high byte)	Hex	Resolution: 1/64/bit, offset: 0. range: 0~100m/s
data #7	delta wheel speed RL (low byte)	Hex	
data #8	delta wheel speed RR (high byte)	Hex	Resolution: 1/64/bit, offset: 0. range: 0~100m/s

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Byte No.	Description	Data Type	value
data #9	delta wheel speed RR (low byte)	Hex	
NOTE: These bytes from data #2 to data #9 are only for RoutineControlType = requestRoutineResults and Supported Control / Routine Status = "finished".			

F.3.6 RID 0x0305- WheelSpeedSensorsTest

Byte No.	Description	Data Type	value
data #1	Supported Control/ Routine Status	Hex	Same as the Comments in the table of F.3.1
data #2	Maximum speed of the FL wheel (high byte)	Hex	Resolution: 1/64/bit, offset: 0. range: 0~100m/s
data #3	Maximum speed of the FL wheel (low byte)	Hex	
data #4	Minimum speed of the FL wheel (high byte)	Hex	Resolution: 1/64/bit, offset: 0. range: 0~100m/s
data #5	Minimum speed of the FL wheel (low byte)	Hex	
data #6	Maximum speed of the FR wheel (high byte)	Hex	Resolution: 1/64/bit, offset: 0. range: 0~100m/s
data #7	Maximum speed of the FR wheel (low byte)	Hex	
data #8	Minimum speed of the FR wheel (high byte)	Hex	Resolution: 1/64/bit, offset: 0. range: 0~100m/s
data #9	Minimum speed of the FR wheel (low byte)	Hex	
data #10	Maximum speed of the RL wheel (high byte)	Hex	Resolution: 1/64/bit, offset: 0. range: 0~100m/s
data #11	Maximum speed of the RL wheel (low byte)	Hex	
data #12	Minimum speed of the RL wheel (high byte)	Hex	Resolution: 1/64/bit, offset: 0. range: 0~100m/s
data #13	Minimum speed of the RL wheel (low byte)	Hex	
data #14	Maximum speed of the RR wheel (high byte)	Hex	Resolution: 1/64/bit, offset: 0. range: 0~100m/s
data #15	Maximum speed of the RR wheel (low byte)	Hex	
data #16	Minimum speed of the RR wheel (high byte)	Hex	Resolution: 1/64/bit, offset: 0. range: 0~100m/s
data #17	Minimum speed of the RR wheel (low byte)	Hex	
NOTE: If any of the wheel speed sensor is invalid, the output value will be 0xFFFF. **NOTE:** These bytes from data #2 to data #17 are only for RoutineControlType = requestRoutineResults and Supported Control / Routine Status = "finished".			

F.3.8 RID 0x0403- Speed Limit Disable/Enable

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Byte No.	Description	Data Type	value
data #1	Supported Control/ Routine Status	Hex	Same as the Comments in the table of F.3.7
NOTE: This request is Disable/Enable the speed Limit (10kph for Extended Mode). **NOTE:** This routineIdentifier 0x0403 cannot be stopped by routineControlType = Stop Control.			

F.3.9 RID 0x0404- Disable Valve Relay Shut down

Byte No.	Description	Data Type	value
data #1	Supported Control/ Routine Status	Hex	Same as the Comments in the table of F.3.7
NOTE: This routineIdentifier 0x0404 cannot be stopped by routineControlType = Stop Control.			

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Annex G (informative) Examples for addressAndLengthFormatIdentifier parameter values

The detailed description about this Annex can be found in Annex G of the document *CN200 General Diagnostic Requirement Specification*.

It is possible for ABS supplier to add the additional definitions in Annex G for CN200 vehicle, if necessary, it is ABS supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

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